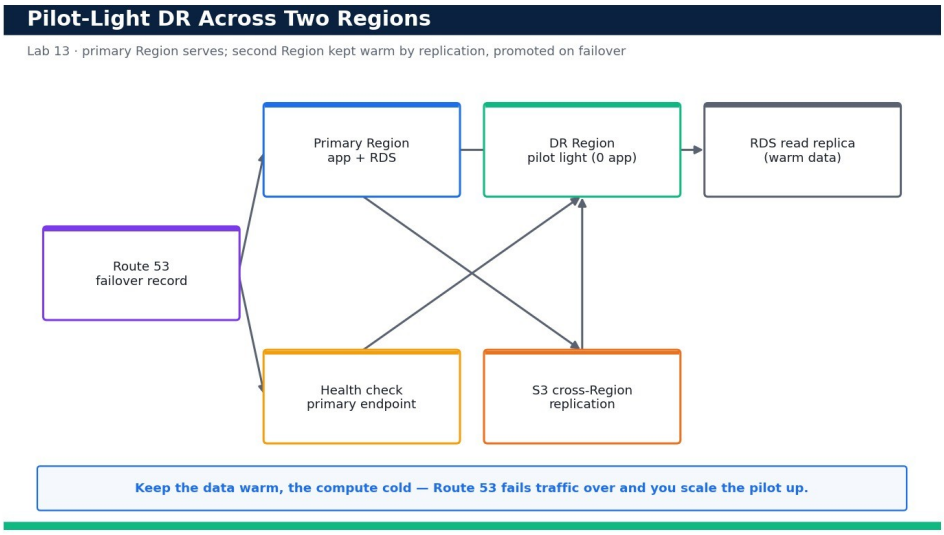
Lab 13 — Select a DR Strategy by RTO/RPO and Implement a Pilot-Light Skeleton

DevOps Engineering on AWS (TGS-2023040474) · Day 3 · Domain 03 Resilient Cloud Solutions · LO3 · K2, K4, A2



Reference architecture — Lab 13

Objective

Compare the four AWS disaster-recovery strategies against RTO/RPO targets, select an appropriate pattern, and implement a pilot-light skeleton in a second Region (LO3, K2, K4, A2).

Learning outcome LO3 · Skills mapped: K2, K4, A2 · Domain 03 · Day 3.

Scenario

Not every workload needs active-active — and not every business can accept a day-long restore. In this lab you complete an RTO/RPO worksheet, compare the four AWS DR strategies (backup & restore, pilot light, warm standby, active-active) against those targets, and then stand up a pilot-light skeleton in a second Region: a minimal network and a scaled-to-zero application tier kept warm by cross-Region data replication, with a Route 53 health check and failover record ready to promote it.

What you'll build

A completed DR selection worksheet plus a deployed pilot-light skeleton in a second Region (minimal VPC, a scaled-to-zero app tier and an S3 cross-Region replication path) fronted by a Route 53 failover record set. Services: Amazon Route 53, Amazon RDS, AWS CloudFormation, Amazon S3 (Cross-Region Replication).

Prerequisites

- An AWS account you control (an AWS Free Tier / sandbox account is sufficient for most steps).

- AWS CLI v2 installed and configured with a least-privilege named profile: aws configure --profile devops (Region e.g. ap-southeast-1).
- Confirm your identity before you start: aws sts get-caller-identity --profile devops.
- Git and a code editor; a modern browser for the AWS Management Console.
- Copy .env.example to .env and fill in YOUR values. .env is git-ignored — never commit real credentials.
- Run every command from this lab folder with your devops profile (append --profile devops or set AWS_PROFILE=devops).
- A Route 53 hosted zone you control (for the failover record set); a private test zone is fine.
- An IAM role (devops-s3-replication-role) that S3 can assume for cross-Region replication.
- A distinct DR Region enabled on the account (e.g. ap-southeast-2).

Cost & safety

Pilot light is deliberately cheap: the DR Region runs only a minimal network, replicated S3 objects and a small RDS instance/replica — no running app fleet. Expect low single-digit US$ for the duration of the lab; the RDS instance and cross-Region replication are the only meaningful line items. Delete both stacks after.

- Use only your own AWS account and the placeholder values in .env.example.
- Never commit real credentials — use IAM roles / the AWS CLI profile and .env placeholders only.
- Run the Clean up section as soon as you finish to stop ongoing charges.

Files in this lab

- dr-plan.md
- pilot-light.yaml
- s3-crr.json
- failover-notes.md
- .env.example
- lab-13-rto-rpo-dr-selection.jpg
- INSTRUCTIONS.pdf

Step-by-step

1. Open the DR worksheet and record the workload's target RTO and RPO before choosing anything.

```
cat dr-plan.md
```

Expected output: The worksheet shows the four strategies with typical RTO/RPO ranges and blank fields for your workload's targets and chosen pattern.

2. Confirm your primary Region and pick a distinct DR Region for the second copy.

```
aws configure get region --profile devops
```

Expected output: Your primary Region prints (e.g. ap-southeast-1); you choose a different DR Region such as ap-southeast-2.

3. Validate the pilot-light template that will run in the DR Region.

```
aws cloudformation validate-template --template-body file://pilot-light.yaml --profile devops
```

Expected output: The command returns the template Description and Parameters (EnvName, DrVpcCidr) with no error.

4. Create an S3 source bucket in the primary Region with versioning enabled (required for replication).

```
aws s3api create-bucket --bucket devops-dr-src-<account-id> --region ap-southeast-1 --create-bucket-configuration LocationConstraint=ap-southeast-1 --profile devops && aws s3api put-bucket-versioning --bucket devops-dr-src-<account-id> --versioning-configuration Status=Enabled --profile devops
```

Expected output: The source bucket is created and versioning reports Status=Enabled.

5. Create the destination bucket in the DR Region, also versioned, and block public access.

```
aws s3api create-bucket --bucket devops-dr-dst-<account-id> --region ap-southeast-2 --create-bucket-configuration LocationConstraint=ap-southeast-2 --profile devops && aws s3api put-public-access-block --bucket devops-dr-dst-<account-id> --public-access-block-configuration BlockPublicAcls=true,IgnorePublicAcls=true,BlockPublicPolicy=true,RestrictPublicBuckets=true --profile devops
```

Expected output: The destination bucket exists in the DR Region with Block Public Access fully enabled.

6. Apply the cross-Region replication configuration so new objects flow to the DR Region automatically.

```
aws s3api put-bucket-replication --bucket devops-dr-src-<account-id> --replication-configuration file://s3-crr.json --profile devops
```

Expected output: The command returns with no error; the source bucket now has a replication rule targeting the DR bucket.

7. Prove replication works by uploading a test object and checking it appears in the DR bucket.

```
aws s3 cp dr-plan.md s3://devops-dr-src-<account-id>/dr-plan.md --profile devops && sleep 30 && aws s3 ls s3://devops-dr-dst-<account-id>/ --profile devops
```

Expected output: After a short delay dr-plan.md is listed in the destination bucket, confirming cross-Region replication (RPO for object data).

8. Deploy the pilot-light skeleton stack in the DR Region (minimal VPC + a scaled-to-zero Auto Scaling group).

```
aws cloudformation deploy --template-file pilot-light.yaml --stack-name dr-pilot --region ap-southeast-2 --parameter-overrides EnvName=dr --capabilities CAPABILITY_IAM --profile devops
```

Expected output: CloudFormation reports stack dr-pilot reaching CREATE_COMPLETE in the DR Region.

9. Confirm the DR app tier is present but running zero instances (pilot light = warm data, cold compute).

```
aws autoscaling describe-auto-scaling-groups --auto-scaling-group-names dr-pilot-asg --region ap-southeast-2 --query 'AutoScalingGroups[0].{Min:MinSize,Desired:DesiredCapacity}' --profile devops
```

Expected output: The DR Auto Scaling group shows MinSize=0 and DesiredCapacity=0 — no compute cost while idle.

10. Create a Route 53 health check against the primary endpoint (an HTTP check on / every 30s) that will drive failover.

```
aws route53 create-health-check --caller-reference dr-$(date +%s) --health-check-config Type=HTTP,ResourcePath=/,FullyQualifiedDomainName=<primary-alb-dns-name>,Port=80,RequestInterval=30,FailureThreshold=3 --profile devops
```

Expected output: A HealthCheck Id is returned; it monitors the primary endpoint over HTTP.

11. Create the primary and secondary failover record sets in your hosted zone.

```
aws route53 change-resource-record-sets --hosted-zone-id <hosted-zone-id> --change-batch file://failover-records.json --profile devops
```

Expected output: A ChangeInfo with Status PENDING is returned; the zone now has a PRIMARY (health-checked) and a SECONDARY failover record.

12. Read failover-notes.md and rehearse the promotion runbook so the team knows exactly what to do on failover.

```
cat failover-notes.md
```

Expected output: The runbook lists the ordered steps: promote/restore data, scale the DR Auto Scaling group up from 0, verify health, and confirm Route 53 has shifted traffic.

13. Rehearse promoting the pilot light: scale the DR Auto Scaling group up from zero to serve traffic.

```
aws autoscaling update-auto-scaling-group --auto-scaling-group-name dr-pilot-asg --min-size 2 --desired-capacity 2 --region ap-southeast-2 --profile devops
```

Expected output: The DR group scales from 0 to 2 InService instances, demonstrating the pilot light can be brought to full capacity.

14. Scale the DR tier back to zero to return to the cheap pilot-light steady state after the rehearsal.

```
aws autoscaling update-auto-scaling-group --auto-scaling-group-name dr-pilot-asg --min-size 0 --desired-capacity 0 --region ap-southeast-2 --profile devops
```

Expected output: The DR group returns to 0 instances, confirming the pilot light is warm-data / cold-compute again.

Validation

The dr-plan.md worksheet records the workload RTO/RPO and a justified strategy choice; the S3 source bucket replicates objects to the DR Region; the dr-pilot stack is CREATE_COMPLETE with a scaled-to-zero Auto Scaling group; and Route 53 has a PRIMARY (health-checked) plus SECONDARY failover record. The rehearsal shows the DR group can scale from 0 to 2 and back.

Expected outcome (Test it)

A test object uploaded to the primary bucket appears in the DR Region bucket (data is warm), and the DR Auto Scaling group can be promoted from 0 to 2 instances and returned to 0 — proving the pilot-light pattern meets the chosen RTO/RPO at minimal steady-state cost.

Evidence to capture

- The completed dr-plan.md worksheet showing the workload RTO/RPO and the justified strategy choice.
- s3 ls of the DR-Region destination bucket showing the replicated dr-plan.md object.
- describe-stacks (DR Region) showing dr-pilot at CREATE_COMPLETE with a MinSize=0 Auto Scaling group.
- change-resource-record-sets output showing the PRIMARY and SECONDARY failover records created.

Troubleshooting

- **put-bucket-replication fails with an IAM error —** Replication needs an IAM role that S3 can assume with s3:GetObjectVersion on the source and s3:ReplicateObject on the destination; create devops-s3-replication-role first and reference its ARN in s3-crr.json.
- **Objects don't replicate —** Both buckets must have versioning enabled and the object must be created AFTER the replication rule; pre-existing objects are not replicated unless you run S3 Batch Replication.
- **Route 53 failover never switches —** The PRIMARY record must have a health check attached and Failover=PRIMARY; the SECONDARY must be Failover=SECONDARY with the same record name and type, and the health check must actually be evaluating your endpoint.

Clean up

Delete everything you created, in dependency order, to avoid ongoing charges:

```
aws route53 change-resource-record-sets --hosted-zone-id <hosted-zone-id> --change-batch file://failover-records-delete.json --profile devops
```

```
aws cloudformation delete-stack --stack-name dr-pilot --region ap-southeast-2 --profile devops
```

```
aws s3 rm s3://devops-dr-src-<account-id> --recursive --profile devops
```

```
aws s3 rm s3://devops-dr-dst-<account-id> --recursive --profile devops
```

```
aws s3api delete-bucket --bucket devops-dr-src-<account-id> --profile devops
```

```
aws s3api delete-bucket --bucket devops-dr-dst-<account-id> --region ap-southeast-2 --profile devops
```

Confirm nothing is left running (no active stacks, instances, dashboards or trails for this lab).

Acceptance criteria

- Every step's Expected output is observed, and the Validation and Test-it checks pass.
- The reference architecture is realised using the editable AWS-native source files in this folder.
- GitHub via AWS CodeConnections is used as the source path where a repository is required (CodeCommit optional).
- All Evidence-to-capture items are collected (screenshots / CLI output).
- No live credentials appear in any committed file; .env is not committed.
- The Clean up step has been run and no billable resources remain.